

Prateek Deshmukh

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Education

SRM Institute of Science and Technology, Kattankulathur

2023-2027

B.Tech Artificial Intelligence

CGPA: 8.98/10

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Networks.

Skills

Programming: Python, TypeScript, C++, SQL, PHP **Web/AI:** React, Next.js, FastAPI, LangChain, Vapi, Pydantic, Node.js **DB/Cloud:** PostgreSQL, SQLite, MongoDB, Redis, GCP, Docker, Linux, Cloudflare **Other:** MCP (Model Context Protocol), Webhooks, RESTful Systems, MVC Architecture, Microservices Architecture

Experience

- **Engineering Intern, Faff** Sep 2025 - Present
 - Architected a core **RecurringTaskService** using a cron-driven pattern to automate task lifecycle management, ensuring seamless context transfer and data integrity across thousands of child task generations over multiple cadences.
 - Developed **LangChain** toolsets allowing LLM-based agents to autonomously interact with the internal ticketing system, enabling automated handling of customer support workflows.
 - Engineered a reliable **ETA management system** that uses automated logic to track deadlines, freeze stalled tasks, and alert teams before breaches occur.
 - Orchestrated a payment gateway migration from Razorpay to **Cashfree**, implementing secure webhook verification, signing checks, and automated **Zoho** GST compliance.
 - Integrated **Vapi AI voice agents** with Google Maps API to automate vendor outreach, optimizing system prompts and transcribers for 25% higher regional communication accuracy.
- **Incoming Software Engineer Intern, NatWest** Expected Summer 2026
 - Selected as 1 of 11 interns from 10,000+ applicants (Top 0.1%) through a multi-stage rigorous competitive evaluation including offline coding and technical interview rounds.
- **Full-Stack SDE Intern, Easylib** May-July 2025
 - Built an internal documentation system and observability platform for 300+ client instances using PHP, CodeIgniter, and MS SQL, focusing on uptime tracking and performance metrics.
 - Developed critical modules and subsystems within the sales desk and library cataloging software, adhering to MVC architecture principles.

Projects

- **Backademia: LLM-Integrated MCP Server** (Python, FastAPI, SQLite)
 - Developed a **Model Context Protocol (MCP)** server that exposes reverse-engineered university portal data (attendance, marks, timetable) as actionable tools for LLMs.
 - Built a RESTful abstraction layer (Backademia API) to handle secure web scraping and session management, enabling agents to execute CRUD operations on institutional data.
 - Implemented robust authentication and rate-limiting to ensure secure, high-resilience data retrieval under heavy loads.
GitHub: <https://github.com/prateekdesh/backademia>
- **Hooked: Webhook Debugging Utility** (Python, Ngrok API)
 - Developed a CLI-based wrapper for Ngrok to streamline webhook integration testing by providing real-time filtering of incoming payloads by headers and JSON keys.
 - Simplified the debugging of third-party API integrations (e.g., Payment Gateways) by isolating specific data formats, reducing webhook debugging time by ~50% during the Cashfree migration.
- **prats.blog CMS** (React, FastAPI, SQLite)
 - Designed a full-stack CMS with a custom rich-text editor, featuring live previews, secure JWT authentication, and optimized image upload workflows.

Achievements

- Awarded 'Most Creative Project' at Epoch Hackathon in Delhi for innovative project design and development.
- Won 1st prize in CodeDay Bengaluru hackathon.